



**DEDAN KIMATHI UNIVERSITY OF TECHNOLOGY PROJECT PROPOSAL FOR
FINAL YEAR STUDY IN B.Sc. BUSINESS INFORMATION TECHNOLOGY**

**POST DIAGNOSIS CHRONIC KIDNEY DISEASE PATIENT'S MANAGEMENT
SYSTEM(TELEMED) -PROJECT PROGRESS PRESENTATION**

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C027-01-2415/2022

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A Project proposal submitted to the department of Information Technology in the School of Computer Science and Information Technology in partial fulfillment of the requirements for the award of the degree of Bachelor of Science in Information Technology at Dedan Kimathi University of Technology.

APRIL,2026.

Acknowledgement

I would like to express my sincere gratitude to my supervisor, **Mr. Mike Muchiri**, for his invaluable guidance, continuous support, and constructive feedback throughout the development of this project. His expertise and dedication greatly contributed to shaping the direction and quality of this work.

I also extend my appreciation to the lecturers and staff of the School of Computer Science and Information Technology at Dedan Kimathi University of Technology for equipping me with the knowledge and skills necessary to undertake this project.

Special thanks go to my family and friends for their encouragement, understanding, and moral support during the entire period of my study. Their patience and motivation played a significant role in the successful completion of this proposal.

Lastly, I acknowledge all researchers and authors whose work has contributed to the development of this project.

Dedication

This project is dedicated to my family, whose unwavering support, encouragement, and sacrifices have been the foundation of my academic journey.

I also dedicate this work to all patients suffering from Chronic Kidney Disease, especially those undergoing dialysis. May this project contribute, even in a small way, toward improving their quality of life through better health management and access to care.

Declaration

I declare that this is my original work and has not been presented in any University for a degree or for any consideration of any certification.

STUDENT: **FREDRICK KARIMAH MWANGOLO**

Sign..... Date.....

This proposal document has been submitted with my approval as the university supervisor

NAME: **Mr MIKE MUCHIRI**

Sign..... Date.....

Abstract

Kidney dialysis patients face ongoing challenges in managing their health due to the complexity of treatment schedules, medication routines, and the need for constant monitoring. This project proposes a digital platform that empowers dialysis patients to actively participate in their care through a structured and AI-assisted health management system. The solution allows users to log daily symptoms and medications, which are then analyzed using machine learning algorithms to detect health trends such as improvement, stability, or deterioration. Based on these insights, the system provides timely recommendations on whether to consult a nephrologist and allows for seamless appointment booking. The platform is designed to be user-friendly, secure, and accessible, with a focus on reducing unmonitored health decline and improving patient-doctor communication. Through this integrated approach, the system aims to enhance the quality of care for dialysis patients, promote early intervention, and support informed clinical decision-making.

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CHAPTER 1

1.0 Introduction

Chronic kidney disease (CKD) is a critical medical condition and ongoing disease process that affects millions of people worldwide, with many dialysis patients regularly monitored with medical interventions in order to maintain optimal quality of life. To manage such an ongoing disease process requires strict adherence to taking medications, monitoring health status, and following up with a healthcare expert like a nephrologist on time. Many dialysis patients are expected to track their symptoms, keep track of medications taken, and recognize subtle changes in their health that may not be major but could negatively impact the health process.

Thereby comes a digital health support system (TeleMed) that enables dialysis patients to self-manage their health daily. This digital health support system harnesses AI-facilitated data(symptoms, signs and medication) analysis, and automated recommendation reports to assist in tracking personal health information, but it promotes early management or persistent of either personal exacerbations, or complications reducing patient outcomes, not to mention visits to the hospital emergency department.

TeleMed aims to:

- i) Reduce preventable complications
- ii) Lower hospital admissions.
- iii) Empower Chronic Kidney Disease patients in managing their own health.

1.1 Background.

Chronic Kidney Disease (CKD) is a progressive and irreversible condition that has become a major global public health concern. According to the World Health Organization, chronic diseases such as CKD are responsible for a significant proportion of mortality worldwide, particularly in low- and middle-income countries. CKD occurs when the kidneys gradually lose their ability to filter waste and excess fluids from the blood, leading to serious health complications such as hypertension, anaemia, cardiovascular disease, and eventually kidney failure. At advanced stages, patients often require renal replacement therapy, including dialysis or kidney transplantation, to sustain life.

Dialysis patients, in particular, face a complex and demanding treatment regimen that requires strict adherence to medication, dietary restrictions, fluid intake monitoring, and frequent medical consultations. Despite these requirements, many patients struggle to consistently monitor and record their daily health status, including symptoms such as fatigue, swelling, shortness of breath, and fluctuations in blood pressure. Failure to track these indicators can result in delayed detection of complications, leading to emergency hospitalizations and reduced quality of life.

Traditionally, patients have relied on manual methods such as paper-based logs to record their symptoms and medication intake. However, these methods are often inefficient, prone to inaccuracies, and difficult to maintain consistently over time. Furthermore, healthcare systems in many regions lack integrated tools that can analyze patient-generated health data in real time, limiting opportunities for early intervention. As a result, valuable health information remains underutilized, and clinical decisions are often made without continuous insight into a patient's day-to-day condition.

The rapid advancement of digital health technologies and artificial intelligence (AI) presents a promising opportunity to address these challenges. AI-driven systems can process large volumes of patient data, identify patterns, and generate actionable insights that support proactive healthcare management. In chronic disease care, such technologies have shown potential in improving patient adherence, enhancing early detection of complications, and reducing unnecessary hospital visits.

Telemedicine platforms have further transformed healthcare delivery by enabling remote monitoring, virtual consultations, and improved communication between patients and healthcare providers. These systems are particularly beneficial for patients with chronic conditions like CKD, who require continuous monitoring but may face barriers to frequent hospital visits, such as distance, cost, or limited healthcare infrastructure.

Despite these advancements, there remains a gap in the availability of specialized digital solutions tailored specifically for dialysis patients. Many existing applications focus on general health tracking and do not incorporate condition-specific analytics, real-time alerts, or seamless integration with nephrology care. This gap highlights the need for a targeted, intelligent system that not only captures patient data but also interprets it meaningfully to support timely medical intervention.

In response to these challenges, this study proposes the development of an AI-assisted digital health support system (TeleMed) designed specifically for chronic kidney disease patients undergoing dialysis. The system aims to facilitate daily health monitoring, provide intelligent analysis of patient-reported data, and enable timely communication with healthcare providers. By leveraging AI and telemedicine, the platform seeks to enhance patient self-management, reduce preventable complications, and ultimately improve overall health outcomes.

1.2 Problem Statement

According to a study published in *BMC Nephrology* (2020), over 40% of chronic Kidney dialysis patients fail to consistently record and monitor their symptoms and medication adherence, increasing the risk of missed early warning signs and the likelihood of post-late detection complications. Symptoms such as fatigue, swelling, abnormal blood pressure, or fluid retention may appear clearly at first, but without structured monitoring, they can escalate into severe medical events requiring emergency care.

Traditional methods of health tracking like paper logs are often inconvenient, inconsistent, and prone to user error, especially for patients managing complex kidney dialysis schedules. This has contributed to research findings that show up to 30% of emergency hospitalizations among chronic kidney dialysis patients are preventable if early and smart monitoring is detected and addressed promptly.

Moreover, existing healthcare systems often lack intelligent tools for analyzing daily patient reported data in real time. Even when information is captured, it typically remains unstructured and under-utilized, offering little support for proactive care. At the same time, scheduling appointments with nephrologists can be delayed or fragmented, resulting in treatment gaps during critical moments.

1.3 Objectives

General Objective:

To develop an AI-assisted platform (TeleMed) for chronic kidney disease patients that enables daily health tracking, trend analysis, and appointment management to support timely medical intervention and improve patient outcomes.

Specific Objectives:

By the end of this project, the system will:

1. **Kidney patient's will be able to key in their daily logs to the system:**
Enable patients undergoing kidney dialysis to log their daily signs, symptoms, and medications through a user-friendly interface for consistent health monitoring.
2. **The system will be able to show the trend of their health Analysis:**
Integrate AI models to analyze patient data over time and detect health trends such as fluid retention or abnormal blood pressure, enabling early detection of complications.
3. **The system will be able to recommend the CKD patients to book appointments with doctors:**
Use AI-generated alerts to notify both patients and doctors when symptoms indicate a need for medical attention, prompting appointment recommendations.
4. **Doctors will have a real-time chat with the kidney patients:**
Provide an integrated appointment booking feature within the platform to allow patients to easily schedule consultations with their nephrologists, ensuring continuity of care.
5. **The system will be able to Analyze the patient's status according to the signs and symptoms logged in and also medication taken:**
Provide an integrated analytical feature within the platform to allow patients to easily monitor their trend, whether improving or not.

1.4 Research Questions

1. How can daily health logs (symptoms and medication) from kidney dialysis patients be effectively captured through a user-friendly digital interface?
2. How can AI be utilized to monitor and analyze patient data trends to detect early signs of health deterioration?
3. To what extent can AI-driven systems accurately recommend medical consultations based on patient-reported data?
4. How can an integrated appointment booking system improve accessibility and efficiency in managing nephrology consultations?

1.4. Scope and Justification.

(i) Scope of the Project

This project is focused on designing and developing a patient's health management/monitoring platform tailored for kidney dialysis patients. The platform will support the following key functionalities:

1. **Daily Symptom and Medication Logging**
The system will enable patients to input daily health information, including physical symptoms (e.g., fatigue, swelling), emotional well-being, and medication intake. The interface will be simple and mobile-friendly to encourage regular usage.
2. **AI-Assisted Health Trend Analysis**
An integrated AI module will analyze patients' logged data over time to detect patterns indicating improvement, stability, or deterioration of health. These insights will be presented in easy-to-understand feedback.
3. **Automated Doctor Recommendations**
Based on AI analysis, the system will notify patients when their health data shows concerning trends that warrant immediate consultation with a nephrologist/Doctor.

4. **Appointment Booking**

The system will allow patients to conveniently book appointments with kidney specialists through the platform, based on urgency levels flagged by the AI.

5. **User Role Management**

The platform will support different user roles, particularly patients and doctors, each with tailored access to data and functionalities relevant to their role.

TeleMed will be designed for individual outpatient use, particularly those receiving dialysis at home or visiting clinics periodically. The backend will be developed using Flask (Python), while MySQL will manage structured data storage. Frontend technologies will include HTML, CSS, and JavaScript with potential for integrating charting libraries for data visualization.

(ii) **Justification of the Project**

1. **Rising Burden of Kidney Disease**

Chronic Kidney Disease (CKD) and end-stage renal disease are on the rise globally, especially in low- and middle-income countries. Dialysis patients often lack consistent self-monitoring, which leads to preventable emergencies and poor treatment outcomes. A digital system that supports proactive care can make a measurable impact.

2. **Gaps in Current Digital Solutions**

While several apps exist for health tracking or chronic illness management, few if any are purpose-built for **dialysis patients**. Most lack tailored symptom fields, nephrologist-specific integration, or real-time AI feedback that empowers patients to understand their condition dynamically.

3. **Preventing Emergency Deterioration**

Dialysis patients are vulnerable to rapid health deterioration if symptoms like shortness of breath, low blood pressure, or swelling are not addressed early. AI-powered analysis can help detect red flags in time and prompt necessary care, reducing hospitalization rates.

4. **Improved Doctor-Patient Engagement**

With real-time symptom logs and intelligent health summaries, nephrologists can make better-informed decisions during appointments. This project will bridge the gap between visits and improve communication.

5. Empowerment Through Data

Patients often feel powerless in managing chronic conditions. A system that not only tracks data but provides meaning through analysis gives patients more control and awareness of their own health journey.

6. Scalability and Public Health Impact

Although starting with a focus on dialysis, the modular nature of the system allows for future expansion to cover other chronic conditions such as diabetes or hypertension. Therefore, the solution has long-term potential to influence digital chronic care in broader contexts.

CHAPTER 2

2. Literature Review

2.1 History of patient's managements application

The development of patient management applications has evolved significantly over the past few decades, beginning with basic electronic health record (EHR) systems in the 1960s. Early systems, such as those developed by the Mayo Clinic and Massachusetts General Hospital, focused primarily on digitizing patient data for internal use. As computing advanced in the 1980s and 1990s, hospital information systems (HIS) began integrating scheduling, billing, and laboratory management, though patient-facing functionalities remained minimal.

The rise of the internet and mobile technology in the early 2000s led to the introduction of Personal Health Records (PHRs), such as Microsoft HealthVault and Google Health. These allowed patients limited access to their health information, but adoption was slow due to usability issues and data fragmentation.

In the 2010s, with the advent of smartphones and wearable devices, patient management apps became more interactive and patient-cantered. Apps like MyChart and Apple Health allowed users to track appointments, medication, and vitals. These platforms emphasized remote monitoring, secure messaging with healthcare providers, and integration with hospital EHR systems, leading to improved patient engagement and chronic disease management.

2.2 Case Study

Digital health technologies are increasingly being leveraged to support the management of chronic illnesses, including Chronic Kidney Disease (CKD). Multiple systems have emerged to assist in patient monitoring, medication adherence, and clinical coordination. However, most solutions lack condition-specific intelligence, real-time analysis, or seamless integration with nephrology care workflows. The following case studies explore existing platforms and their limitations in the context of kidney dialysis patient support.

Case Study 1: General-Purpose mHealth Apps

MyChart, Care Clinic, Medisafe

These applications allow users to log symptoms, medications, and lifestyle data. They include features such as medication reminders, symptom logs, and progress reports. While broadly useful, they are:

- ❖ Designed for general chronic conditions, not tailored to dialysis-specific needs.
- ❖ Lack specialist integration, such as nephrologist access.
- ❖ Do not offer real-time AI analysis or early-warning feedback mechanisms.

Gap: No personalized AI-driven trend monitoring or kidney-specific triage capability.

Case Study 2: Kidney-Focused Health Platforms

RenalTracker, Kaiser Permanente's Renal Program, MyKidneyCoach

These systems target CKD patients more directly.

For instance:

- ❖ RenalTracker provides dietary education and symptom journaling but is passive and does not alert patients to urgent issues.
- ❖ MyKidneyCoach includes symptom tracking but lacks machine learning and predictive features.

Gap: These systems lack dynamic AI-based analysis and real-time alerts that inform both patients and clinicians about health deterioration.

Case Study 3: AI in Chronic Disease Management

IBM Watson Health, Google Health Predictive Models

These technologies demonstrate the potential of AI in chronic care. Use cases include:

- ❖ Predicting hospital readmissions,

- ❖ Detecting medication non-adherence,
- ❖ Monitoring risk in heart failure and diabetes.

A 2020 study in *BMC Nephrology* used machine learning to predict mortality in dialysis patients using historical clinical data.

Gap: These models are typically used in clinical settings by professionals not integrated into everyday patient self-monitoring systems.

Case Study 4: Appointment Scheduling Tools

Zocdoc, HealthTap

These platforms offer user-friendly interfaces for booking doctor appointments. However:

- ❖ They are general-purpose and not connected to a patient's symptom trends.
- ❖ They do not use AI to recommend or prioritize urgent nephrology consultations.

Gap:

Lack of symptom-to-action integration i.e., no AI-based recommendation that bridges symptom logging with urgent bookings.

2.4 Proposed Architecture

TeleMed is a digital health management platform focused on supporting kidney dialysis patients. It is designed to collect daily patient inputs, analyze trends using AI, and facilitate interactions with nephrologists through appointment bookings. The system emphasizes usability, timely health insights, and integration of intelligent feedback mechanisms.

System Components:

1. Frontend (User Interface):

- Technology: HTML5, CSS3, JavaScript, Bootstrap
- Purpose: Allows patients to log daily health data, view AI health analysis results, receive recommendations, and book doctor appointments.

2. Backend (Application Logic):

- Technology: Python with Flask Framework
- Responsibilities:
 - ❖ Receives and processes form data from the frontend.
 - ❖ Triggers AI analysis on user data.
 - ❖ Stores and retrieves patient data from the database.
 - ❖ Sends alerts or recommendations based on AI results.

3. Database:

- Technology: MySQL
- Usage: Stores user profiles, medication logs, symptom records, AI analysis history, appointment data, and doctor information.

4. AI Module:

- ❖ Tools: LLMs : Gemini/Open AI

Functionality:

- Analyze trends in patient-reported symptoms and medication adherence.
- Identify risk patterns indicating stable, improving, or deteriorating health.
- Provide interpretable output and feedback messages

5. Appointment Management Module:

- ❖ Built into the backend.
- ❖ Enables patients to view nephrologist availability and book sessions.
- ❖ Sends confirmation messages.

CHAPTER 3

3. Methodology.

3.0 Introduction

This chapter presents the methodology used in the development of TeleMed.

It outlines the research design, data collection methods, software development model, and phased approach to design, build, test, and deploy the system. Ethical considerations and progress tracking mechanisms are also discussed.

3.1 Research Design

The research employed a Descriptive Research Design, chosen for its suitability in understanding user needs through observational and survey data. This design was effective in revealing patient behaviour, pain points in health tracking, and how technology can improve chronic care management.

Justification:

- ❖ Detailed User Insight: Helps uncover the health routines, medication adherence, and challenges faced by dialysis patients.
- ❖ Low-Cost & Accessible: Ideal for academic research and constrained project budgets.
- ❖ Data Interpretability: Simplifies analysis of structured and semi-structured health data.
- ❖ Feature Alignment: Guided the design of the AI module, patient dashboard, and booking system.

3.2 Data Collection Methods

Both primary and secondary data collection methods were utilized.

3.2.1 Secondary Data

Secondary research involved reviewing:

- ❖ Peer-reviewed journals on chronic disease management and AI in healthcare.
- ❖ Mobile health (mHealth) applications such as MyKidneyCoach, CareDx AlloCare, and MyChart.
- ❖ Clinical guidelines from National Kidney Foundation (NKF) and WHO.

This provided insights into current solutions and best practices.

3.2.2 Primary Data

Primary data was obtained from:

- ❖ Clinician Feedback: Informal interviews with nephrologists and clinical medics to understand monitoring gaps and desired patient self-reporting features.

3.3 Software Development Model

Agile methodology is adopted to facilitate iterative development, user feedback integration, and continuous enhancement of features critical to dialysis patients.

Agile Justification:

- ❖ Allows for rapid updates based on clinician/patient feedback.
- ❖ Sprint-based cycles help prioritize modules like AI analysis and appointment booking.
- ❖ Enables progressive testing of AI feedback accuracy and patient interaction flow.

3.4 Development Phases

Phase 1: Requirement Gathering

- ❖ Requirements were gathered from survey insights, user interviews, and clinical consultation(DeKUT Medical Centre).
- ❖ Core features identified:
 - Daily logging of medication and symptoms
 - AI trend analysis
 - Appointment booking system
- ❖ AI use cases were documented, e.g., detecting health deterioration trends and recommending medical reviews.

Phase 2: Design

- ❖ AI response flows were designed to simulate conversations such as:
 - “You have reported fatigue for 3 days. Would you like to book a doctor’s appointment?”
- ❖ A MySQL database schema was designed to store health logs, patient profiles, and AI outputs.
- ❖ System architecture included backend (Flask), frontend (HTML/CSS/JS), and AI module integration.

Phase 3: Development

- ❖ Frontend: Developed using Bootstrap, HTML, and JavaScript for responsiveness and accessibility and UI.
- ❖ Backend: Built with Flask (Python) due to its lightweight nature and ease of integrating ML models.
- ❖ Database: MySQL used for managing user profiles, logs, and appointment records.
- ❖ AI Module: Trained on synthetic and open-source chronic illness symptom datasets. The model predicts health status trends (e.g., worsening/stable/improving).

Phase 4: Testing

- ❖ Unit testing was applied to individual components like symptom logging, appointment scheduling, and AI prediction outputs.
- ❖ User Acceptance Testing (UAT): Conducted with dialysis patients and clinicians to validate AI recommendations and system usability.
- ❖ Bug Fixes & Iteration: Changes were made based on user feedback to improve prompt clarity and interface flow.

Phase 5: Deployment

- ❖ The system was deployed on a local server and hosted web interface, accessible on both desktop and mobile devices practicality was confirmed using NGROK.
- ❖ Patients interacted with the system through real-life simulation and dummy health logs.
- ❖ Feedback from early deployment helped refine the AI threshold for when a consultation is recommended.

3.5 Ethical Considerations

- ❖ **Consent:** Participants involved in interviews and testing provided verbal or written consent.
- ❖ **AI Transparency:** Patients are informed that AI suggestions are non-diagnostic and advisory in nature.
- ❖ **Security:** No sensitive health credentials or diagnostic decisions are stored or made by the system.

CHAPTER 4

4. Implementation.

4.1 Resources

TeleMed utilizes a combination of software and hardware resources essential for designing, developing, testing, and deploying a functional web-based health management application. These resources are categorized as follows:

4.1.1 Software Resources

1. Programming Languages / Frameworks

I will use the following programming languages and frameworks to implement the project objectives

- ❖ **Back-end:**

- Python 3.11 with the Flask web framework
This will be used for the backend design and framing of the UI/UX for responsive of web and mobile interface.

- ❖ **Front-end:**

- HTML5, CSS3, JavaScript with Bootstrap 5
This will be used for the design and implementation of the UI/UX.

- ❖ **AI/ML Integration:**

- Gemini 2.5 will be used for the analysis of the daily logs and medication taken from the users (Patients) and it will be resourceful for the recommendation generation for faster doctor act.

- ❖ **APIs:**

- Flask RESTful AP. This will be ideal for backend/frontend communication for easy communication between the client and server side.

2. Database Management

- ❖ **Main Database:**

- MySQL: this will be used for managing user profiles, health logs, medical history, and appointment records and the recommendations generated by the nephrologists

3. AI Tools and Libraries

❖ **Trend Analysis Engine:**

Gemini 2.5 will be used for the analysis of the daily logs and medication taken from the users (Patients) and it will be resourceful for the recommendation generation for faster doctor act

❖ **Natural Language Interaction.**

- A rule-based engine mimicking chatbot-like prompts (e.g., recommending medical reviews when risk levels rise).

4. Development Tools

❖ **Version Control:**

- Git & GitHub This will be used for collaborative code management and backup when a disaster occurs

❖ **IDE:**

- Visual Studio Code with Python, Flask, and MySQL extensions

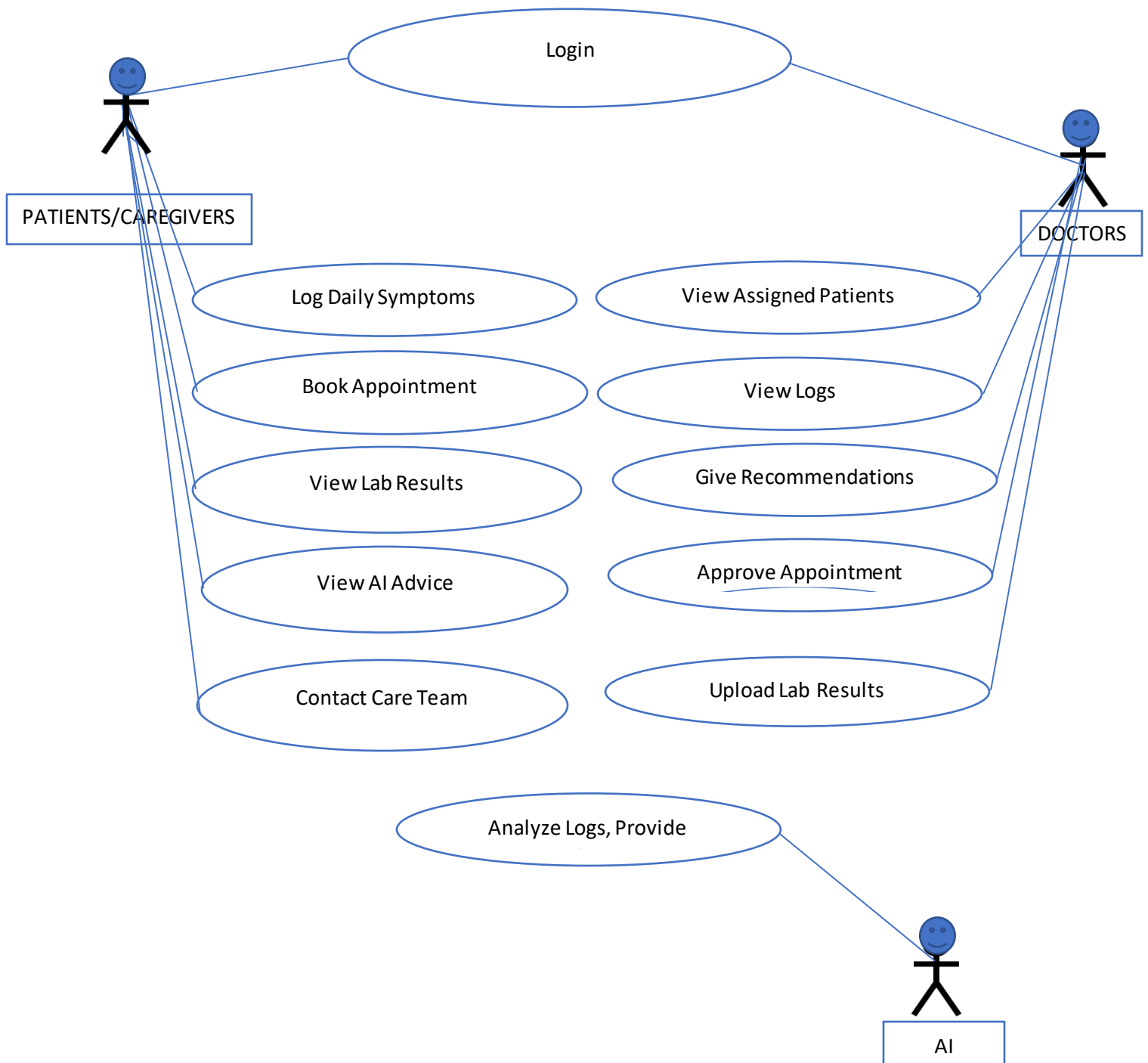
4.1.2 Hardware Resources

❖ **Development Machine:**

- Laptop with Intel i5 processor, 8GB RAM, and 256GB SSD, capable of running Python applications, database servers (MySQL), and AI integration.

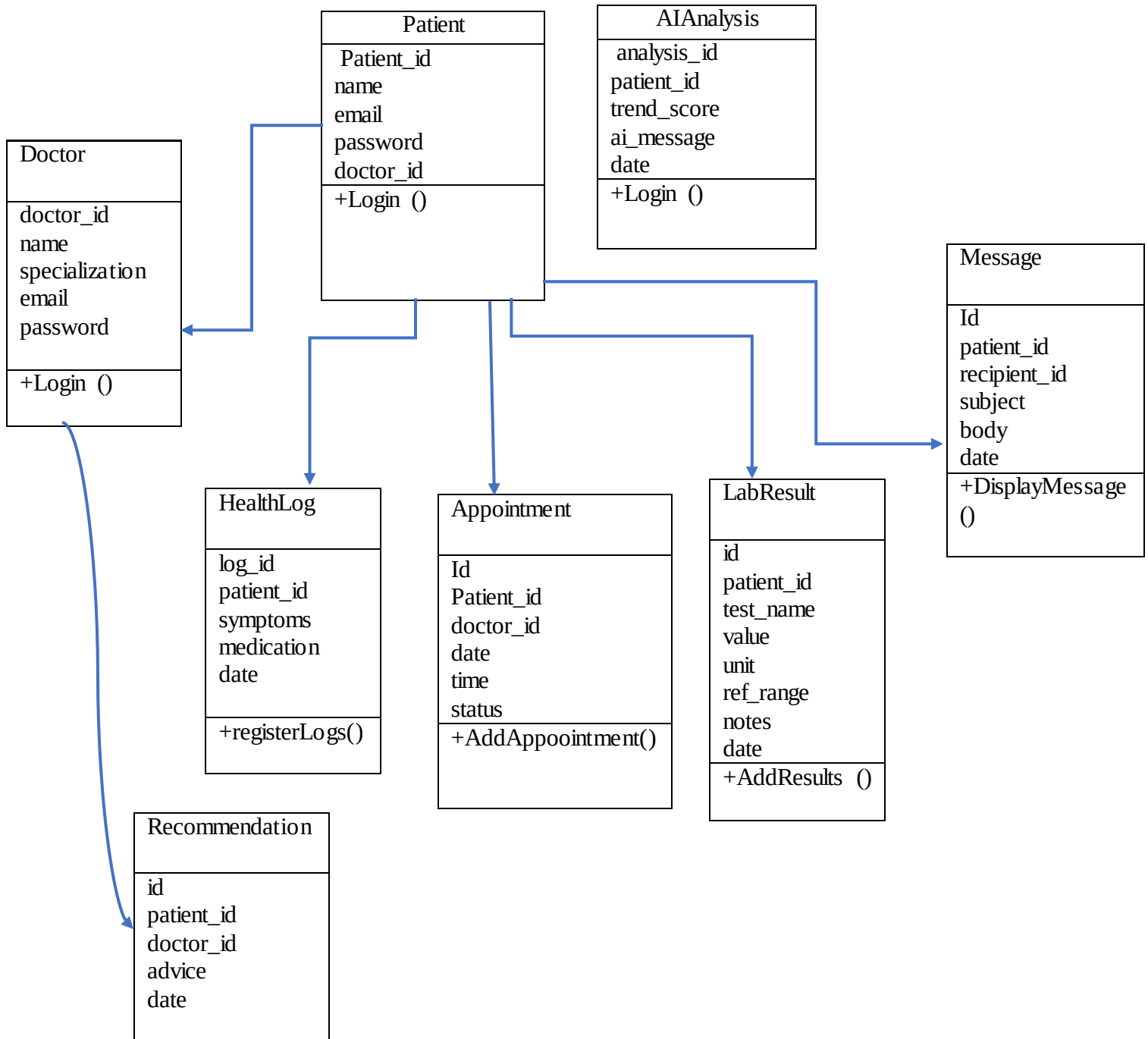
4.2 Design.

4.2.1 UML Process Design



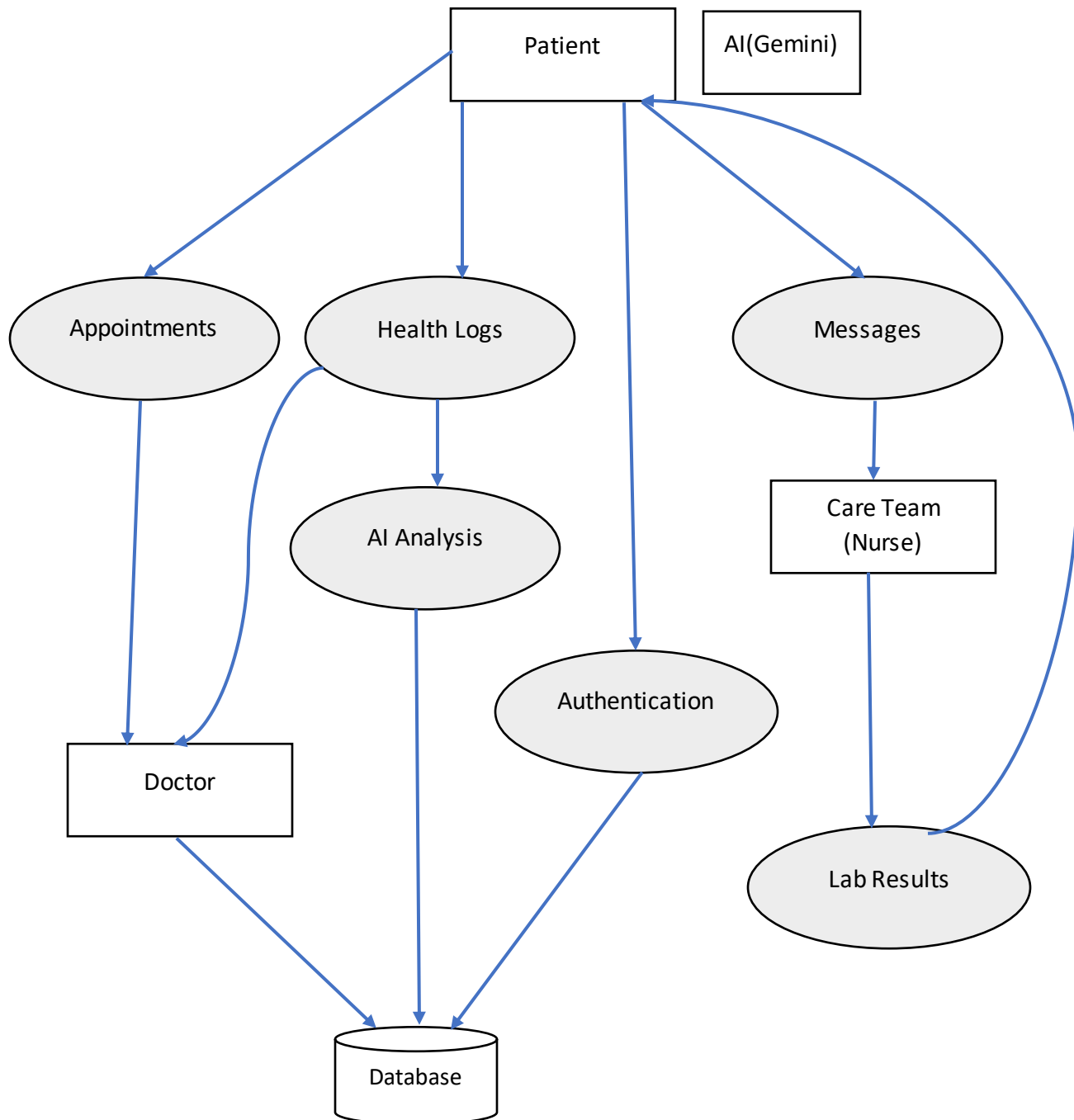
4.2.2 Class diagrams

The class diagram models the main entities of the TeleMed system such as Patient, Doctor, Health Log, Appointment, and AIAnalysis. It shows their attributes and relationships, ensuring clear data organization.



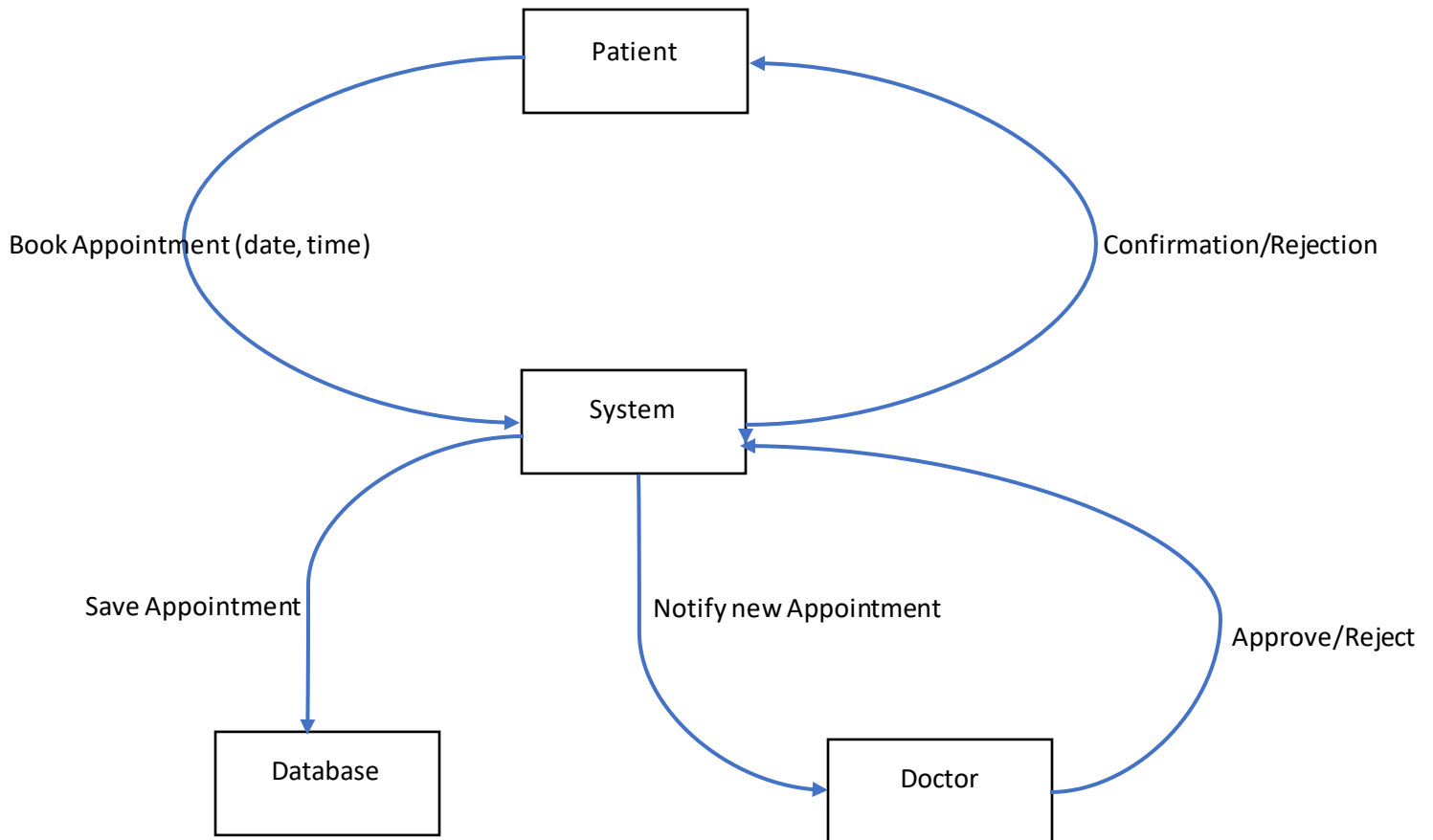
4.2.3 DFD (Logical Flow)

The data flow diagram represents the logical flow of data between external entities (Patient, Doctor, Nurse, AI) and processes (Authentication, Health Logs, Appointments, Labs, Messages, AI Analysis).



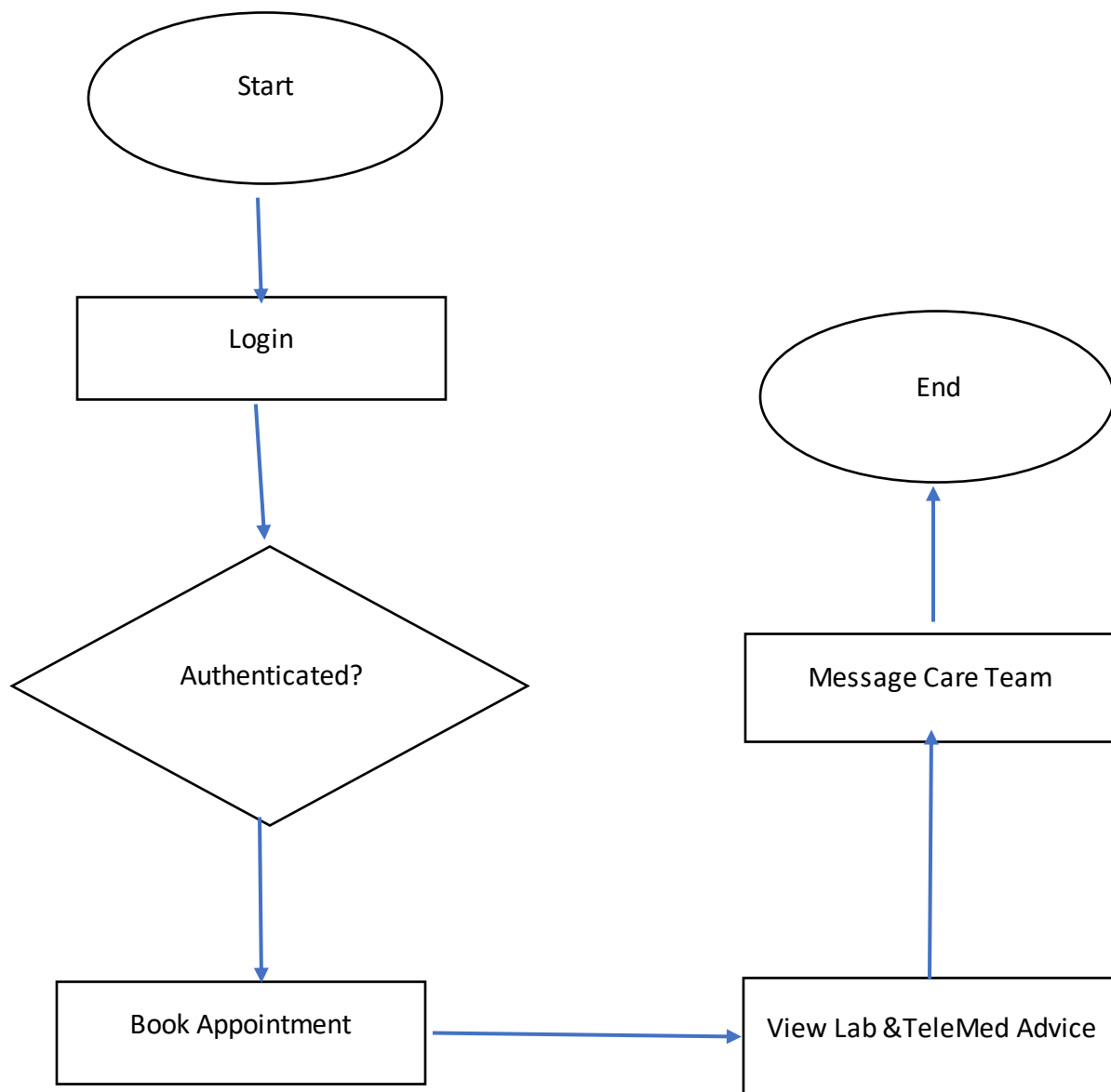
4.2.4 Sequence Diagram

The sequence diagram demonstrates the appointment booking process, showing the interaction between the Patient, System, Database, and Doctor, including booking, saving, notification, and approval



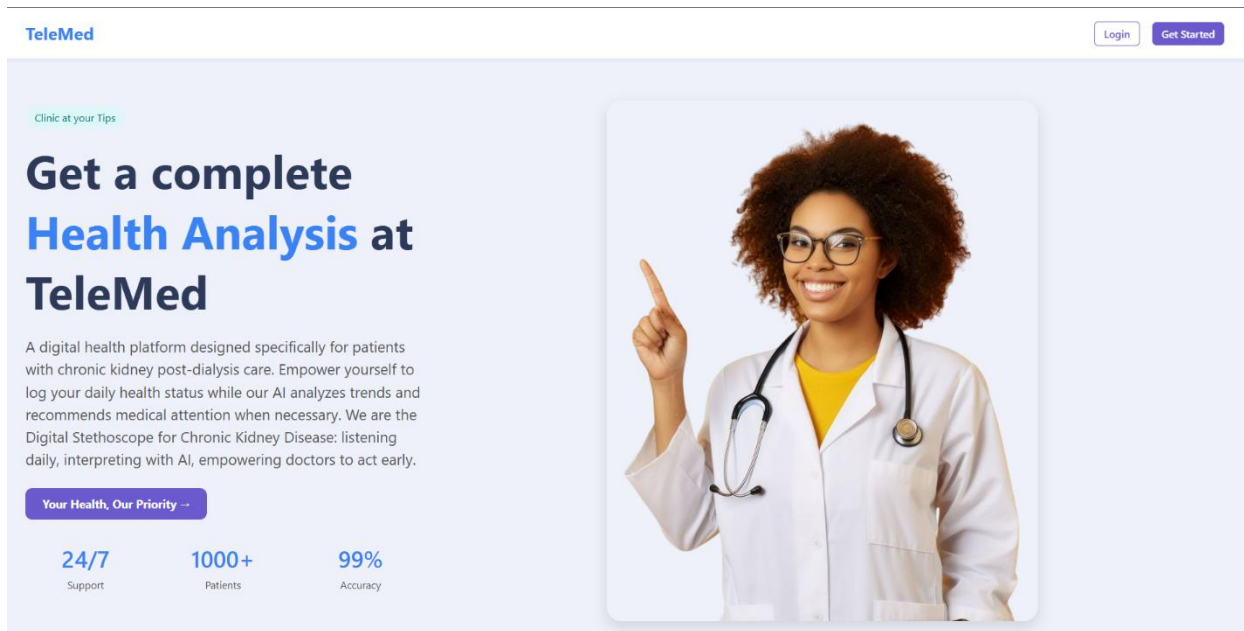
4.2.5 Activity Diagram

The activity diagram illustrates the workflow of a patient within the system. It includes login, logging symptoms, booking appointments, viewing lab results and AI advice, and messaging the care team.



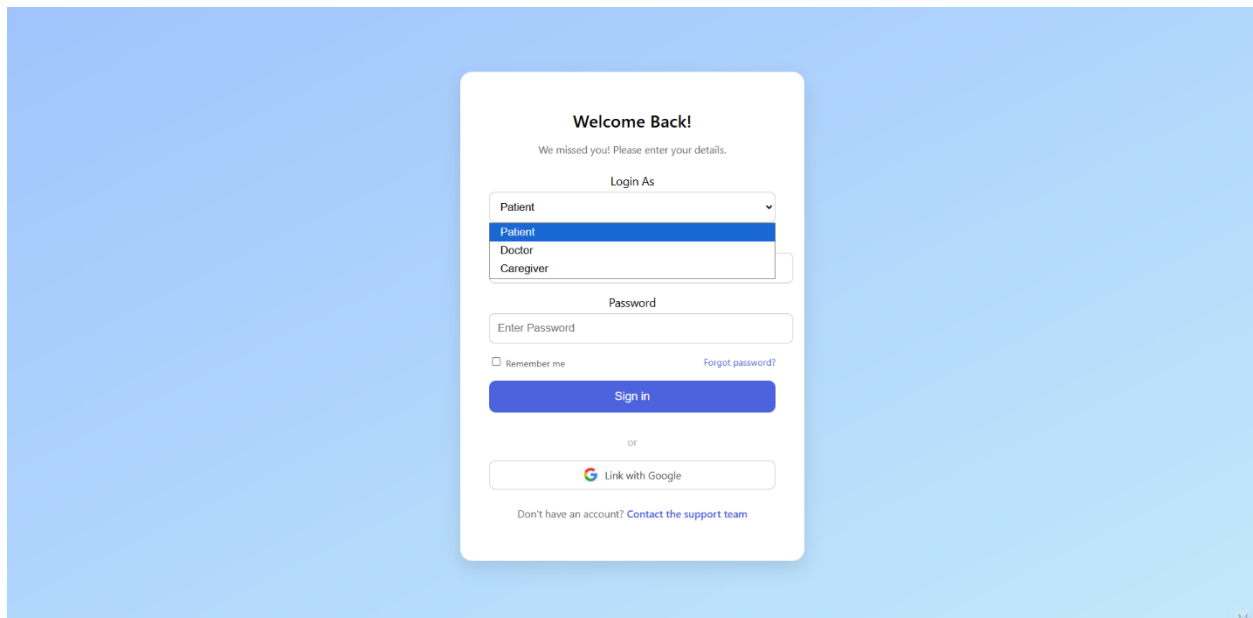
4.2 Object Achieved/Output

Home page of the system



Login interface of the system.

(One can log in as Doctor, Patient or Caregiver)



Patient's Dashboard (i.e. can log daily logs, book appointment, see his/her health trend)

Welcome, Wendy Abdul!

Track your health status and stay connected with your care team

Log Today's Status

Symptoms

Describe any symptoms...

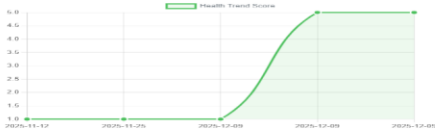
Medications Taken

List medications taken...

Submit Log

Health Trend Analysis

Health Trend Overview



Telemed Analysis:

Condition: Worsening Advice: Assess for intradialytic hypotension or neurological causes for headache/dizziness. Review fluid status.

Nephrologist's Recommendations

Dr. James Otieno (2025-12-09 10:46:38)

Let meet tomorrow

Dr. James Otieno (2025-11-25 15:28:05)

Lets have an impromptu appointment

Dr. James Otieno (2025-09-27 03:39:37)

I advice you to persist on fluids

Dr. James Otieno (2025-09-18 15:52:22)

Lets meet on the agreed appointment

Dr. James Otieno (2025-09-08 22:08:30)

Your Last 5 Logs

2025-12-09 10:41:44

Symptoms: dizziness, and an acute headache

Medications: No medicine taken

2025-12-09 09:54:55

Symptoms: dizziness, and an acute headache

Medications: No medication taken

2025-12-09 09:54:12

Symptoms: Symptoms: Swelling in ankles persists, dialysis effective Appetite fair, blood pressure within normal range

Medications: No medicine taken

Book Appointment

Assigned Doctor: Dr. Dr. James Otieno

Specialization: Internal Medicine

Date

dd/mm/yyyy

December 2025

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Confirm Appointment

33 | Page

Doctor's Dashboard (i.e. can monitor daily logs, approve/ reject appointment, give recommendations based on and get AI recommendation.)

Doctor Dashboard
Welcome, Dr. James Otieno
Logout

My Patients

Name	Email	Recent Logs	Recommendation	AI Suggestion
Wendy Abdul	wendy@gmail	2025-12-09 10:41:44 – dizziness, and an acute headache (No medicine taken) 2025-12-09 09:54:55 – dizziness, and an acute headache (No medication taken) 2025-12-09 09:54:12 – Symptoms: Swelling in ankles persists, dialysis effective Appetite fair, blood pressure within normal range (No medicine taken) 2025-11-25 15:23:45 – Mild fatigue in the morning, slight swelling in ankles, no chest pain (Sevelamer carbonate 800mg – with meals (taken)) 2025-11-12 13:56:00 – Mild fatigue in the morning, slight swelling in ankles, no chest pain (Sevelamer carbonate 800mg – with meals (taken))	Recommend	AI Suggestion
Wanyo	wanyo@gmail.com	2025-11-14 12:29:10 – I'm now feeling better (N/A) 2025-11-14 12:26:53 – Headache , swollen ankles (Panadol and water) 2025-11-12 14:27:56 – I've been feeling very tired and weak lately, like I have no energy even after resting. I sometimes feel dizzy or lightheaded (ibuprofen) 2025-11-12 14:01:42 – head ache running stomach pain in the abdomen (panadol mara moja) 2025-09-18 15:58:01 – I can't walk well my abdomen is itchy (No medicine taken)	Recommend	AI Suggestion

		effective Appetite fair, blood pressure within normal range (No medicine taken) 2025-11-25 15:23:45 – Mild fatigue in the morning, slight swelling in ankles, no chest pain (Sevelamer carbonate 800mg – with meals (taken)) 2025-11-12 13:56:00 – Mild fatigue in the morning, slight swelling in ankles, no chest pain (Sevelamer carbonate 800mg – with meals (taken))		
Wanyo	wanyo@gmail.com	2025-11-14 12:29:10 – I'm now feeling better (N/A) 2025-11-14 12:26:53 – Headache , swollen ankles (Panadol and water) 2025-11-12 14:27:56 – I've been feeling very tired and weak lately, like I have no energy even after resting. I sometimes feel dizzy or lightheaded (ibuprofen) 2025-11-12 14:01:42 – head ache running stomach pain in the abdomen (panadol mara moja) 2025-09-18 15:58:01 – I can't walk well my abdomen is itchy (No medicine taken)	Recommend	AI Suggestion

Pending Appointments

ID	Date	Patient	Action
7	2025-09-16 00:00:00	Wendy Abdul	Accept Reject
9	2025-09-28 00:00:00	Wendy Abdul	Accept Reject
10	2025-11-14 00:00:00	Wanyo	Accept Reject
11	2025-11-26 00:00:00	Wendy Abdul	Accept Reject
12	2025-12-10 00:00:00	Wendy Abdul	Accept Reject

CHAPTER 5

System Coding, Testing and Deployment.

5.1 Introduction

This chapter details the implementation phase of the TeleMed Patient Management System. It articulates the transition from system design to a fully functional software application. The chapter covers three critical stages: System Coding (constructing the software architecture), System Testing (verifying functionality and reliability), and Deployment (making the system accessible to users). The primary objective was to ensure the system is secure, user-friendly, and capable of handling real-time medical data efficiently.

5.2 System Coding and Implementation

System coding involved translating the logical design and architectural specifications into executable code. The development followed the Model-View-Controller (MVC) architectural pattern, ensuring a clean separation between data management, user interface, and business logic.

5.2.1 Development Environment and Tools

The selection of tools was driven by the need for scalability, data security, and rapid development.

- **Frontend (The View):** HTML5, CSS3, JavaScript, and Bootstrap (for mobile responsiveness).
- **Backend (The Controller):** Python 3.10 using the **Flask Microframework**.
- **Database (The Model):** MySQL for relational data management.
- **AI Agent & Data Analysis:** Gemini 2.5 lite for the symptom trend analysis model.
- **Version Control:** Git for source code management and GitHub for remote repository storage.
- **Virtual Environment:** venv was used to manage dependencies and isolate the project environment.

5.2.2 Backend Implementation (Logic Layer)

The backend was built using Flask, chosen for its flexibility and compatibility with machine learning libraries. Key implementation details include:

- **Modular Routing:** The application was structured using Flask Blueprints. This separated distinct logic handles: such as auth_routes (login/register), doctor_routes, and patient_routes to improving code readability and maintainability.
- **Authentication & Security:** User sessions were managed using Flask-Login. Passwords were hashed using Werkzeug security functions before storage to ensure data privacy.
- **AI Integration:** The AI agent (Gemini 2.5 flash lite) for symptom trend analysis was integrated directly into the backend routes. When a patient submits a health log, the Python backend preprocesses the data, passes it through the agent, and returns the prediction to the doctor's dashboard.
- **Notification Logic:** A polling mechanism was implemented to handle in-dashboard notifications, alerting users to appointment updates instantly on the cancellation of the appointments booked just in case there are.

5.2.3 Frontend Implementation (Presentation Layer)

The frontend focused on accessibility and "Mobile-First" design, acknowledging that many patients would access the system via smartphones.

- **Responsive Interfaces:**

Bootstrap grid systems were utilized to ensure the Patient and Doctor dashboards render correctly on devices of all screen sizes.

- **Dynamic Rendering:**

The Jinja2 templating engine was used to dynamically populate HTML pages with data passed from the backend (e.g., displaying a specific patient's medical history).

- **Interactive Elements:**

JavaScript was employed for client-side interactions, such as modal pop-ups for appointment booking and dynamic form validation.

5.2.4 Database Implementation (Data Layer)

MySQL was configured to handle structured medical data. The implementation focused on data integrity:

- **Schema Design:**

Tables were created for Users, HealthLogs, Appointments, and Notifications.

- **Relationships:**

Foreign keys were strictly enforced to maintain relationships (e.g., a HealthLog must belong to a valid PatientID).

- **Normalization:**

The database was normalized to the third normal form (3NF) to reduce data redundancy.

CHAPTER 6

System Documentation.

5.1 Introduction

System documentation is a vital component of the software development lifecycle, ensuring the longevity, maintainability, and usability of the application. This chapter outlines the documentation created for the TeleMed Patient Management System. It is divided into two main categories:

1. **Technical Documentation:** Directed at system administrators and developers, detailing how the system is constructed, installed, deployed and configured.
2. **User Manual:** Directed at the end-users (Patients, caregivers and Doctors), providing step-by-step instructions on how to utilize the system's features effectively.

The objective of this documentation is to minimize the learning curve for new users and provide a clear reference for future maintenance or system upgrades.

6.2 Technical Documentation

The technical documentation describes the system's internal mechanics, intended for developers who may wish to modify, debug, or deploy the application in the future

6.2.1 System Requirements

To run the TeleMed system efficiently, the hosting environment must meet the following minimum specifications:

Hardware Requirements (Server Side):

- **Processor:** Intel Core i5 or equivalent (2.0 GHz or higher).
- **RAM:** Minimum 4GB (8GB recommended for handling AI processing).
- **Storage:** 500MB free space for application code and database storage.

Software Requirements:

- **Operating System:** Windows 10/11, macOS, or Linux (Ubuntu recommended for deployment).
- **Runtime Environment:** Python 3.10 or higher.
- **Database:** MySQL Server 8.0.
- **Browser:** Google Chrome, Mozilla Firefox, or Microsoft Edge (latest versions).

6.2.2 Installation and Configuration Guide

The following steps outline the procedure for setting up the development environment:

1. **Repository Cloning:**
 - Clone the source code from the GitHub repository: `git clone https://github.com/username/telemed-system.git`
2. **Virtual Environment Setup:**
 - Create a virtual environment to isolate dependencies: `python -m venv venv`
 - Activate the environment:
 - Windows: `venv\Scripts\activate`
 - Mac/Linux: `source venv/bin/activate`
3. **Dependency Installation:**
 - Install the required Python packages (Flask, Pandas, MySQL-Connector, etc.): `pip install -r requirements.txt`
4. **Database Configuration:**
 - Import the provided `telemed_db.sql` file into your MySQL server.
 - Update the database connection string in `config.py` with your local credentials (host, user, password).
5. **Running the Application:**
 - Start the Flask server: `python app.py`
 - Access the system via the browser at: `http://localhost:5000`

6.2.3 Code Structure

The application follows a modular Flask structure to ensure organization:

- `app.py`: The entry point of the application.
- `/static`: Contains CSS stylesheets, JavaScript files, and images.
- `/templates`: Contains Jinja2 HTML files for the user interface (e.g., `dashboard.html`, `login.html`).

- /routes: Contains Python scripts handling backend logic (e.g., auth_routes.py, patient_routes.py).
- /analyze_logs: Contains the AI agent (Gemini 2.5 flash lite) for trend analysis.
- .
- **Database:** MySQL Server 8.0.
- **Browser:** Google Chrome, Mozilla Firefox, or Microsoft Edge (latest versions).

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6.3 User Manual

The User Manual serves as a guide for the non-technical users of the system. It is designed to be simple, using non-technical language to explain how to perform day-to-day tasks.

6.3.1 Getting Started

- **Accessing the System:** Open a web browser and navigate to the system URL (or the Ngrok link provided during the pilot phase).
- **Landing Page:** Users are greeted with a welcome screen giving them the option to "Login" or "Register."

6.3.2 Patient Module Guide

This section guides patients on managing their health records.

1. **Registration:**
 - Click "Register as Patient."
 - Fill in personal details (Name, Age, Email, Password).
 - Submit the form to create an account.
2. **Logging Health Symptoms:**
 - Navigate to the "Health Log" tab on the dashboard.
 - Select current symptoms from the dropdown menu and enter vital signs (e.g., Temperature, Heart Rate).

- Click "Submit." The system will display an immediate AI-generated trend analysis (e.g., "Condition Improving").

3. **Booking Appointments:**

- Go to "Book Appointment."
- Select a doctor and a preferred time slot.
- Click "Confirm." The status will show as "Pending" until the doctor accepts.

6.3.3 Doctor Module Guide

This section assists doctors in managing patient care.

1. **Dashboard Overview:**

- Upon login, the doctor sees a summary of "Total Patients," "Pending Appointments," and "Critical Alerts."

2. **Managing Appointments:**

- Navigate to the "Appointments" section.
- View pending requests. Click "Accept" to confirm or "Reschedule" to suggest a new time.

3. **Viewing Patient History:**

- Click on a patient's name to view their profile.
- Access historical health logs and AI analysis charts to aid in diagnosis.

6.3.4 Troubleshooting

- **"Invalid Login" Error:** Ensure the Caps Lock is off and that the email address is spelled correctly.
- **System Not Loading:** Check your internet connection. If the issue persists, contact the administrator.
- **Password Reset:** If a password is forgotten, use the "Forgot Password" link on the login page to initiate a recovery process.

CHAPTER 7

Conclusion and Recommendation.

7.1 Conclusion

The development of the TeleMed Patient Management System was driven by the need to bridge the gap between healthcare providers and patients through technology. By integrating automated health logging, AI-driven symptom trend analysis, and a streamlined appointment scheduling system, the project successfully addressed the core challenges of remote patient monitoring.

The implementation phase proved that using a Flask-based backend coupled with a MySQL relational database provides a robust and secure environment for sensitive medical data. The inclusion of an AI component demonstrated that machine learning can significantly assist doctors by providing data-driven insights before a consultation even begins.

Ultimately, the system met its primary goal: creating a centralized, accessible platform that improves the efficiency of healthcare delivery and empowers patients to take an active role in managing their well-being.

7.2 Recommendations

Based on the development process and the feedback received during the testing phase, the following recommendations are made to enhance the system's effectiveness:

- **End-User Training:**

To maximize the benefits of the AI trend analysis, brief training sessions or video tutorials should be provided to healthcare staff to help them interpret the system's predictive insights accurately.

- **Enhanced Security Protocols:**

For a full-scale clinical rollout, it is recommended to implement Two-Factor Authentication (2FA) and end-to-end encryption for data in transit to ensure compliance with global health data standards (such as HIPAA or GDPR).

- **Data Diversification:**

The AI model's accuracy can be further improved by training it on a wider variety of localized datasets to ensure it accounts for regional health trends and demographics.

- **Stakeholder Feedback Loop:**

Continuous feedback from both doctors and patients should be integrated into the maintenance cycle to ensure the user interface remains intuitive and addresses real-world pain points.

7.3 Future Work

While the current version of TeleMed is a functional and effective tool, the following areas have been identified for future development:

- **Native Mobile Applications:**

Developing dedicated Android and iOS applications using the existing Flask API would allow for push notifications and better integration with mobile-specific health sensors.

- **Wearable Device Integration:**

Future iterations could include synchronization with wearable tech (e.g., Apple Watch, Fitbit) to automate the logging of vitals like heart rate, sleep patterns, and daily steps.

- **Teleconference Integration:**

Incorporating a built-in video consultation feature (using WebRTC) would allow doctors to conduct virtual check-ups directly within the platform.

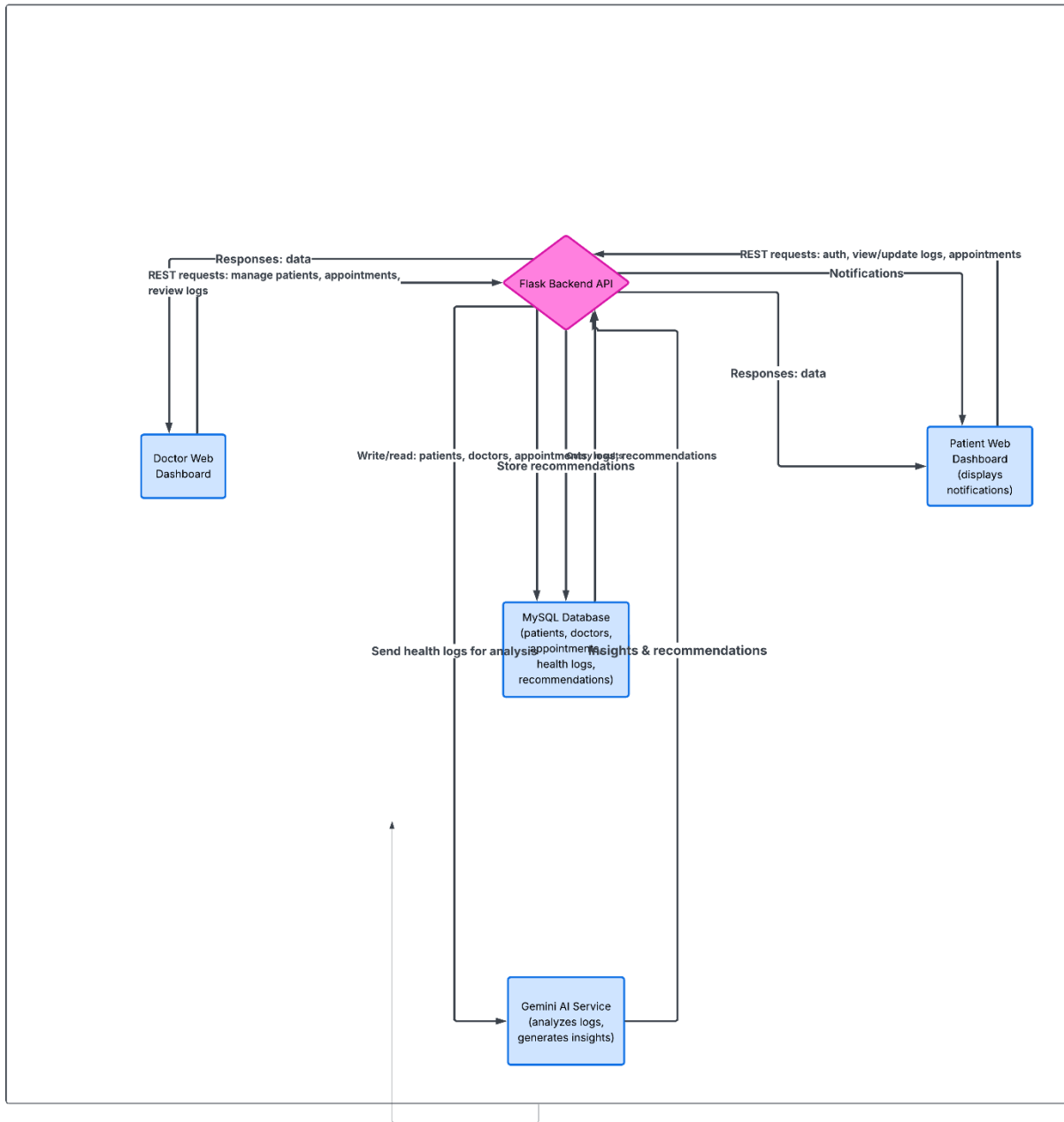
- **Pharmacy & Lab Integration:**

Expanding the system to include digital prescriptions and direct links to laboratory management systems would create a truly end-to-end healthcare ecosystem.

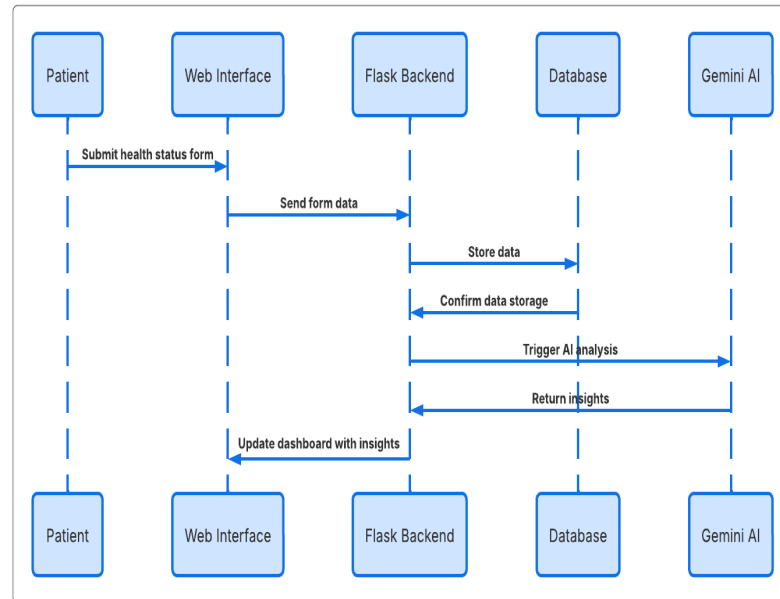
Scheduling

Duration	July	August	October	November	November
Task					
Project proposal writing and Presentation					
Research					
Analysis					
Design					
Development					
Deployment					

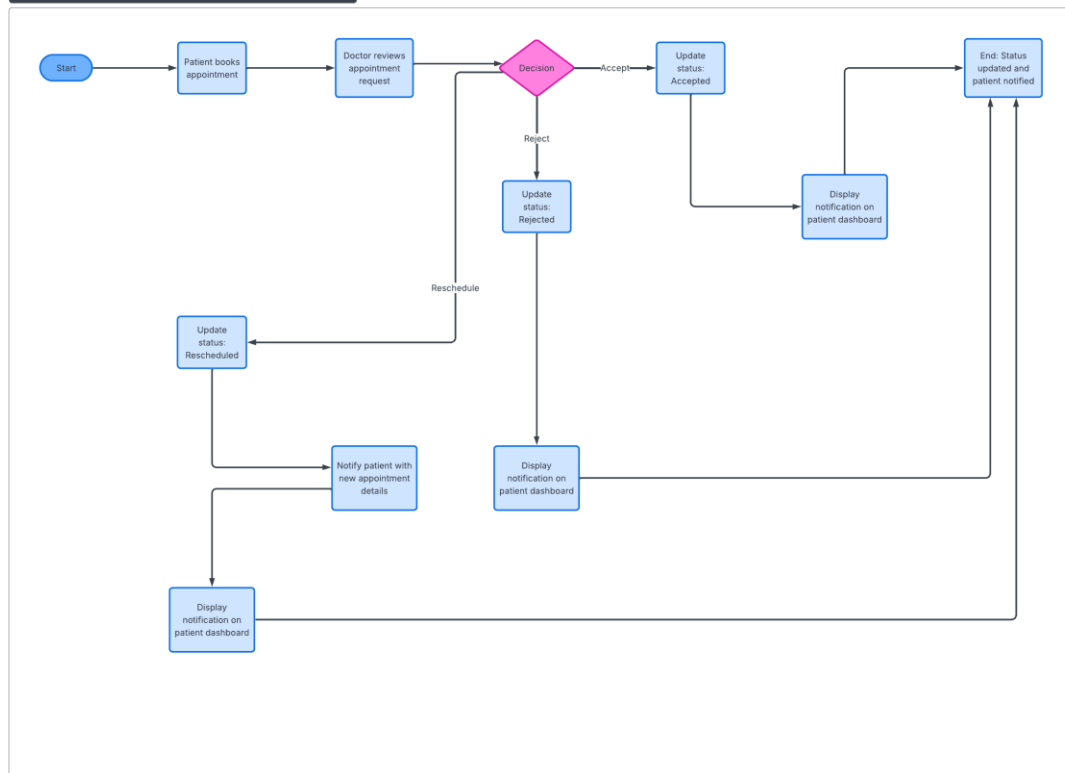
TeleMed Kidney Dialysis System Architecture



"TeleMed Health Status Logging Sequence"



"TeleMed Appointment Management Activity"



4.1 REFERENCES

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4.1 APPENDIX

A.1 Login Page

The login page allows users to access the system based on their roles ie

- ❖ Doctor
- ❖ Patient
- ❖ Caregiver
- ❖ Admin

A.2 Patient Dashboard

- ❖ The patient's dashboard enables CKD patients to:
- ❖ Log daily symptoms
- ❖ Record medication taken
- ❖ View AI health analysis
- ❖ Monitor Trend charts

A.4 Appointment Booking Page

The lab results section displays:

- ❖ Appointment Scheduling
- ❖ Real time Status
- ❖ Book consultations with assigned Doctors

A.5 Laboratory Page

The lab section Displays

- ❖ Test names
- ❖ Results value
- ❖ Reference ranges
- ❖ Clinical notes
- ❖ Test status